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Management Research Based on the Paradigm of the Design Sciences: The Quest for Field-Tested and Grounded Technological Rules

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ABSTRACT Academic management research has a serious utilization problem. In this field mainstream research tends to be description-driven, based on the paradigm of the 'explanatory sciences', like physics and sociology, and resulting in what may be called Organization Theory. This article argues that the relevance problem can be mitigated if such research were to be complemented with prescription-driven research, based on the paradigm of the 'design sciences', like Medicine and Engineering, and resulting in what may be called Management Theory. The typical research products in Management Theory would be 'field-tested and grounded technological rules'. The nature of such rules is discussed as well as the research strategies producing them.

INTRODUCTION

In virtually all academic disciplines research is undertaken to create valid and reliable knowledge to pass on to students and to share with other interested parties. This also applies to research done in Business Schools. As most students of Business Schools regard them as Professional Schools and aspire to management careers outside academia, the use of research results for the practice of management should be a major issue.

Yet there are serious doubts about the actual relevance of present-day management theory as developed by the academic community. As far back as 1982, Beyer and Trice remarked, 'Recently . . . scholars have expressed concern about why organisational research is not more widely used' (Beyer and Trice, 1982, p. 591). In their study of American management education, Porter and McKibbon (1988) remark that the business world is, generally speaking, ignoring the research

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coming from Business Schools. Management fads, scorned by academics, seem to have much more impact on management actions (Abrahamson, 1996). Voicing similar concerns in his Presidential Address to the American Academy of Management, Hambrick (1994) sketched a dismal picture of the Academy's impact and concluded that it might have mattered to the world of organizations and business, but that it did not.

Essentially, the solution Hambrick proposes is to improve the presentation of academic management research results to the outside world, in order to 'open up the incestuous, closed loop of the Academy's conferences' (Hambrick, 1994, p. 13). Improving researcher practitioner communication certainly will help. My thesis, however, is that the relevance problem of academic management theory is not only caused by poor presentation but also by its nature. Compare Kurt Lewin's well-known adage, 'nothing is quite so practical as a good theory' (Lewin, 1945, p. 129). Assuming that 'good' means something like 'scientifically valid and reliable' rather than 'practical', this article intends to qualify that adage: all good theories are practical, but some are more practical than others.

The nature of the products of a given research programme (in Lakatos' (1991) sense) or 'school of thought' (McKinley et al., 1999) is largely determined by its research paradigm. By 'research paradigm' I mean the combination of research questions asked, the research methodologies allowed to answer them and the nature of the pursued research products. Most academic research in management is based on the notion that the mission of all science[1] is to understand, i.e. to describe, explain and possibly predict (see e.g. Emory, 1985; Nagel, 1979). Some even state that 'the essence of science is explanation by law' (Seth and Zinkhan, 1991, p. 35). But, also many non-positivists hold that the mission of all science is to create shared understanding, i.e. understanding of a certain phenomenon shared between the researcher and an informed audience, his scientific community (Peirce, 1960).

However, understanding a problem is only halfway to solving it. The second step is to develop and test (alternative) solutions. Understanding the sources of resistance to certain organizational changes, still leaves undone the task of developing sound change programmes. Understanding the reasons for delays in New Product Development still leaves undone the task of developing effective product development systems. Understanding the changes on certain markets still leaves undone the task of developing successful strategies. Thus, in management one needs next to description-driven research programmes also prescription-driven research ones in order to develop research products which can be used in designing solutions for management problems. By this I do *not* mean the actual application of scientific knowledge to solve a specific managerial problem – this is the domain of practitioners – but the development of scientific knowledge to solve a class of managerial problems, in other words, the development of abstract knowledge. Nor is it a plea to develop recipes, but rather a plea for the develop-

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ment of *field-tested and grounded technological rules to be used as design exemplars* of managerial problem solving. What is meant by such rules will be discussed in detail below.

The classics in our field like Taylor, Fayol and Barnard, did not shrink from prescription, but the subsequent scientization of our field has greatly diminished the academic respectability of prescriptions. In this article I use analogies with various other disciplines, like Medicine and Engineering – which I call 'design sciences' – to analyse the nature of the academic research products used in problem solving and the nature of academic research strategies producing these products. I discuss the differences between these research products and strategies and those of the 'explanatory sciences', like physics and sociology and I use these analogies to show that prescription-driven academic research can indeed claim academic respectability. In applying these ideas to our field, I propose to make a distinction between on the one hand *Organization Theory*, resulting from description-driven research, having an explanatory nature and to be used largely in a conceptual way, and on the other *Management Theory*, resulting from prescription-driven research and to be used largely in an instrumental way to design solutions for management problems. I then discuss for the subfield of Management Theory its typical research products and research strategies and I discuss how the development of such Management Theory may help to mitigate the utilization problem of academic management research.

THE UTILIZATION PROBLEM OF ACADEMIC MANAGEMENT RESEARCH

In the social sciences the utilization problem is a well-known one (see e.g. the 1982 and 1983 Special Issues of *Administrative Science Quarterly* on the utilization of social science; Beyer (1982) gives an introduction to both). In management research it is sometimes seen as a dilemma: the rigour-relevance dilemma (see e.g. Argyris and Schön, 1991, p. 85). Management theory is either scientifically proven, but then too reductionistic and hence too broad or too trivial to be of much practical relevance, or relevant to practice, but then lacking sufficient rigorous justification. March and Sutton (1997) remark that in other disciplines this dilemma is sometimes solved by separating the two contexts. However, in and around Business Schools 'the soldiers of organisational performance and the priests of research purity often occupy not only the same halls but also the same bodies' (March and Sutton, 1997, p. 703).

In the natural sciences the utilization problem is of quite a different order. Some tension may exist between basic and applied research, including a possible difference in social status and competition for research resources. In general, however, two very effective partnerships are in place. There is the one between the natural sciences and applied fields like Medicine and Engineering and the other between

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researchers in any given field and the professionals of that field (often occupying the same bodies).

In the field of management the utilization problem is both a well-discussed and thorny issue (Miner, 1984; Whitley, 1988). Managers and academics have different frames of reference with respect to management knowledge (Shrivastava and Mitroff, 1984) and play different language games (Astley and Zammuto, 1992). Academic researchers, aspiring to relevant research products, have to operate within two reputation systems (March and Sutton, 1997; Whitley, 1988): the academic reputation system, which rewards rigorous research; and the professional reputation system, which rewards relevant research outcomes and the professional training of prospective managers.

The priorities given to each system vary over time and sometimes resemble a pendulum. Prior to the Ford and Carnegie Foundation reports on American Business Schools (Gordon and Howell, 1959; Pierson and Others, 1959), priority was given to professional training, to the professional reputation system. At the time, the academic community regarded the field more or less as a practice-based craft. This was largely caused by the scant attention given to descriptive research and to the justification of the prescriptions given. Examples are the prescriptions on rational decision-making in organizations and the concept of top managers as rational, long-range planners.

The above-mentioned reports started a process of scientization, resulting in a 'New Look' for the American Business Schools (Schlossman et al., 1987; see also Whitley, 1988), which also had a strong impact elsewhere. This process of scientization could have followed the example of the breakthrough of the engineering sciences in the nineteenth century: their assimilation of the laws and especially of the methods of the natural sciences to test and further develop solutions, transformed them from practice-based crafts to solid sciences. However, the Business Schools followed this example only halfway through. The insights and methods of the natural sciences and especially those of the social sciences were used to develop description-driven research programmes, while the interest in prescription atrophied. Gaining recognition in the academic reputation system became the main emphasis. In time, this led to reactions like the *Harvard Business Review* papers 'The myth of the well-educated manager' (Sterling Livingstone, 1971) or 'Managing our way to economic decline' (Hayes and Abernathy, 1980).

Tensions between the two reputation systems are not typically American. In The Netherlands, for instance, the field of business economics has known fierce debates between the Amsterdam School, primarily interested in the academic reputation system and the Rotterdam School, more interested in the professional reputation system (Van Baalen, 1995).[2] In France one finds a somewhat similar competition between the more professional Grandes Ecoles and the more academic Universities. More recently in Britain, Tranfield and Starkey (1998) and Starkey and Madan (2001) voiced concerns with respect to the relevance and application of

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management research results, advocating more emphasis on mode 2 knowledge production (using the distinction Gibbons et al. (1994) make between mode 1 knowledge production, predominantly driven by academic concerns, and mode 2 knowledge production, trans-disciplinary with intensive interaction between knowledge production and knowledge dissemination and application). Such tensions between academia and professional application may have stimulated the idea of the rigour-relevance dilemma (Argyris and Schön, 1991, p. 85). However, this article is not based on the idea that there is a dilemma (in which case no satisfactory solution exists), but rather on the idea of Pettigrew's primary double hurdle: management theory should meet criteria of scholarly quality *and* managerial relevance (Pettigrew, 1997). Understandably, academics tend to worry more about the scholarly hurdle than about the relevance hurdle.[3]

The utilization problem discussed here concerns management theory as developed by the academic community. There is an abundance of management literature, which is widely read by managers but which does not meet scientific standards. This type of literature is dubbed 'Heathrow-literature' by Burrell (1989) or, more kindly, 'Literature on Principles' (of management) by Whitley (1988) or 'Normative' by Miller et al. (1997). There are *craftsman-like* publications, based predominantly on first-hand experience (or on the experience of people one knows), that have a generalization problem: what can be learnt from this experience for other contexts? Then there are *metaphysical* publications by management gurus (nomen est omen), that have a justification problem: on which observations and which logical reasoning are the recommendations based? These publications may perhaps succeed in taking Pettigrew's second hurdle, but fail at his first one. Improving the utility of academic management theory should make it a powerful competitor for these two types of management literature.

Beyer and Trice (1982) give an in-depth analysis of the process of utilizing management research results. Among others, they distinguish between adoption, i.e. the decision by decision-makers within the user system to use certain research results, and implementation, i.e. the actual use of the research results by members of the user system. Another distinction is between instrumental and conceptual use of scientific knowledge (Pelz, 1978).[4] Instrumental use involves acting on research results in specific and direct ways, while in case of conceptual use the results are used for general enlightenment on the subject in question.

The primary interest of this article is the *adoption* of management research results and management theory for *instrumental use*. The problems of subsequent implementation are not very different from other problems of organizational change and implementation and are well-researched. I agree with Beyer and Trice's statement, 'The predominant use of organizational research probably occurs through gradual seepages into organizations of new ideas, metaphors, and rationales for explaining human behaviour' (Beyer and Trice, 1982, p. 615). Such conceptual use of management research results is indeed an important outcome

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(see e.g. Kilduff and Kelemen, 2001, on the power of such conceptual use). However, I fear that academic management research will retain its utilization problem if this remains its only ambition. If Business Schools would take their mission as Professional Schools seriously, they should not only do research with the classical objective of developing knowledge for general understanding and conceptual use, but should also do research with the more ambitious objective of developing knowledge for instrumental use.

THE PARADIGM OF THE DESIGN SCIENCES

The thesis of this article is that a major inhibition for adopting academic management theory for instrumental use lies in the nature of this theory. This nature is strongly influenced by the paradigm used for developing the theory. Kuhn's (1962) term 'paradigm' as used by him has many different meanings. Here it is used in its sociological sense (Masterman, 1970): a system of 'scientific habits' used by a group of scientists for the solution of scientific problems. More specifically, as stated in the introduction, by research paradigm I mean the combination of research questions asked, the research methodologies allowed to answer these questions and the nature of the pursued research products.

Formal, Explanatory and Design Sciences

On the basis of the paradigms used, I distinguish three categories of scientific disciplines:

(1) The *formal* sciences, such as philosophy and mathematics.
(2) The *explanatory* sciences, such as the natural sciences and major sections of the social sciences.
(3) The *design* sciences, such as the engineering sciences, medical science and modern psychotherapy.

The formal sciences are 'empirically void'. Their mission is to build systems of propositions whose main test is their internal logical consistency.

The mission of an explanatory science is to describe, explain and possibly predict observable phenomena within its field. Research should lead to 'true' propositions, i.e. propositions which are accepted by the scientific forum as true on the basis of the evidence provided. The typical research product of an explanatory science is the causal model, preferably expressed in quantitative terms.

The mission of a design science is to develop knowledge for the design and realization of artefacts, i.e. to solve *construction problems*, or to be used in the improvement of the performance of existing entities, i.e. to solve *improvement problems*. Architects and civil engineers deal predominantly with construction problems

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while medical doctors and psychotherapists deal mainly with improvement problems. Research aims at developing knowledge and its application should lead to the intended results. I use the term 'design sciences' because the ultimate objective of research in these sciences is to develop valid and reliable knowledge to be used in designing solutions to problems. I prefer to avoid the term 'applied sciences', as this term suggests that the mission of these sciences is merely to apply the basic laws of the explanatory sciences, thus disregarding the impressive body of knowledge developed by the design sciences themselves.

The idea to distinguish between explanatory and design sciences is strongly inspired by Simon's *The Sciences of the Artificial* (Simon, 1969).[5] Much research within the design sciences is based on the explanatory paradigm, i.e. research aimed at describing, explaining and predicting in order to understand the setting of construction or improvement problems and to know the properties of the 'materials' to be used. However, understanding alone is not enough. The ultimate mission is to develop design knowledge, i.e. *knowledge that can be used in designing solutions to problems* in the field in question. It is important to teach a civil engineer subjects like physics and mechanics, but in designing a bridge he or she needs the design knowledge developed by his or her discipline, like for instance the properties of different types of bridges. In the same way a medical doctor should have a working knowledge of physics and biology, but for medical problem solving he or she predominantly uses the results of the clinical research of his/her own discipline.

In English the term 'science' is often equated with 'natural science', which leads to the idea that the mission of *all* sciences is to merely describe, explain and predict and that such descriptive knowledge is sufficient for practitioners to solve their problems. Science, then, occupies Schön's (1983) 'high ground of theory', while practitioners operate 'in the swamp of practice'. In the present article the focus is on the development of design knowledge, which occupies the middle ground between descriptive theory and actual application. In design sciences like Engineering and Medicine a significant part of this knowledge is produced by academic research, which scores high in both the academic and in the professional reputation system.

Design Knowledge and its Instrumental Use

A design science does not develop knowledge for the layman, but rather for the professionals in its field. This means that design knowledge is to be applied by individuals who have received formal education in that field.

A professional, such as a medical doctor, architect, psychotherapist, mechanical engineer, lawyer or accountant, can be defined as a member of a well-defined group who solves real-world problems with the help of skills, creativity and scientific design knowledge (Abbott, 1996; Becher, 1999; Freidson, 1973; Klegon, 1978; Schön, 1983).

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Each time a professional sets out to solve a unique and specific problem for a client, or in conjunction with a client, he or she does so by using the *problem solving cycle*, also called the regulative cycle (Van Strien, 1997). This cycle consists roughly of: defining the problem out of its 'messy' context (Schön's (1983) 'naming and framing'), planning the intervention (diagnosis, design of alternative solutions, selection), applying the intervention and evaluating.

The essence of professional work is designing, planning an action in advance or during the action ('reflection-in-action'; Schön, 1983). The outcome of this process is a design, which can be defined as a representation of a system or process to be realized. In general, a professional will make three designs (Van Aken, 1994): an *object-design*, the design of the intervention or of the artefact; a *realization-design*, i.e. the plan for the implementation of the intervention or for the actual building of the artefact; and a *process-design*, i.e. the professional's own plan for the problem solving cycle, or, put differently, the method to be used to design the solution to the problem. For instance, in mechanical engineering the object design is the set of drawings of a designed machine, the realization design the set of instructions for the workshop on how to build that machine, and the process design the plan for the design process itself (involving steps like definition of requirements, sketching, making the outline design, making detail designs).

The term 'design science' is used here to indicate that the mission of (academic) research in such a field is to develop scientific knowledge to support the design of interventions or artefacts by professionals and to emphasise its knowledge-orientation: a design-science is not concerned with action itself, but with *knowledge to be used in designing solutions, to be followed by design-based action*.

The Repertoire of Design Knowledge

In order to be able to make these designs, professionals have a repertoire of design knowledge at their disposal (Schön, 1983). This includes their own experience and that of their teachers, and the body of scientific design knowledge of their design science acquired during their training and continuing education. This design knowledge is general, i.e. valid for *classes* of cases. The problem of the professional, however, is always unique and specific. Therefore, general knowledge must be translated to the unique and specific case at hand. In this way, lawyers use case-law from similar cases when dealing with their specific case and doctors use general descriptions of symptoms, diseases and therapies applied previously, when designing a therapy for a specific patient.

Design-repertoires contain three types of design knowledge, according to the three types of designs discussed above. The repertoire of a professional typically contains predominantly object knowledge, i.e. knowledge on the settings and properties of the artefacts or interventions to be designed. For a mechanical engineer

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this may be the properties of different types of bearings and for a medical doctor the effects of alternative therapies for a given disease. It may also contain realization knowledge, e.g. knowledge on manufacturing technologies for a mechanical engineer and knowledge on various types of surgery for a surgeon. Finally, a design repertoire typically contains only a fairly limited amount of explicit process knowledge, i.e. knowledge on how to tackle the actual design process itself. Most professionals obtain their process knowledge in a craftsman-like manner, i.e. by their own experience and by imitating their teachers and peers. Process-knowledge tends to remain largely tacit; professionals often find it difficult to express their approach to design problems.[6]

The design repertoires of well-educated and experienced professionals contain a large variety of knowledge. Within each of the three types of design knowledge discussed above, prescriptions are an important category. The logic of a prescription is 'if you want to achieve Y in situation Z, then perform action X'. There are algorithmic prescriptions which operate like a recipe and which typically have a quantitative format and whose effects can be proven on the basis of observations through deterministic or statistical generalization.[7] However, many prescriptions in a design science are of a heuristic nature. They can rather be described as 'if you want to achieve Y in situation Z, then something like action X will help'. 'Something like action X', means that the prescription is to be used as a *design exemplar*. A design exemplar is a general prescription which has to be translated to the specific problem at hand; in solving that problem, one has to design a specific variant of that design exemplar. For instance, in civil engineering a suspension bridge is one out of several design exemplars an engineer can use to design his or her bridge for his or her specific situation.

In many professions such prescriptions often have a qualitative format. The underlying logic is as stated above, but the actual formulation can use various formats. An example of an algorithmic technological rule is: in order to cure disorder Y, you follow a course of treatment consisting of taking 0.3 milligrams of medicine X during 14 days. An example of a heuristic technological rule is: in order to cure disorder Y, you follow a course of treatment consisting of rest, exercising and a fat-free diet. The indeterminate nature of a heuristic technological rule makes it impossible to prove its effects conclusively, but it can be tested in context, which in turn can lead to sufficient supporting evidence.

Tested and Grounded Technological Rules

In the explanatory sciences, the research object is an '*explanandum*' (Van Strien, 1997) and the typical research product is the causal model: one or more dependent variables are explained in terms of one or more independent variables. Knowledge about the values of these variables can be used to predict the behaviour of the dependent variables.

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In the design sciences the research object is a '*mutandum*' (Van Strien, 1997); these sciences are not too much interested in what *is*, but more in what *can be*. The typical research product is the prescription discussed above, or in terms of Bunge's (1967) philosophy of technology, the *technological rule*. A technological rule is 'an instruction to perform a finite number of acts in a given order and with a given aim' (Bunge, 1967, p. 132). In this article I define a technological rule as *a chunk of general knowledge, linking an intervention or artefact with a desired outcome or performance in a certain field of application*. The 'general' in this definition means that it is not a specific prescription for a specific situation, but a general prescription for a class of problems. On the other hand a technological rule is not a *universal law*, its use being limited to a certain field of application.

Mankind has a long tradition of developing technological rules. Primitive societies not only developed technological rules to manufacture artefacts, but also to make rain, to pacify the gods, to increase fertility, to avoid natural disasters, and so on. So both the know-how of making a bow and arrow and the rain dance are examples of ancient (heuristic) technological rules. A major breakthrough occurred with the systematic testing of technological rules. A tested technological rule is one whose effectiveness has been systematically tested within the context of its intended use. The system of interest is treated as a black box, but under certain conditions specific interventions give the desired results (deterministically or stochastically). Traditional Chinese medicine is an example of a system of very powerful, tested technological rules.

The real breakthrough came when tested technological rules could be grounded on scientific knowledge (Bunge, 1967), including law-like relationships from the natural sciences. For instance, one can design an aeroplane wing on the basis of tested, technological (black box) rules, but such wings can be designed much more efficiently on the basis of tested *and* grounded technological rules, grounded on the laws and insights of aerodynamics and mechanics. The stunning progress of the design sciences since the first Industrial Revolution is based on the effective partnership between the explanatory natural sciences and the design sciences, which leads to systems of tested and grounded technological rules. Whereas the typical research product of the explanatory sciences is the causal model, the typical research product of the modern design sciences is the tested and grounded technological rule.

Clinical Research and the Reflective Cycle

If the tested and grounded technological rule is the typical research product of a design science, the typical research strategy is clinical research, i.e. research on the performance of interventions or artefacts, executed within the context of intended use.

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The causal model of the explanatory sciences is developed, typically, within a closed system (like a laboratory) in order to exclude (or control) the influences on the dependent variables from other sources than the independent variables of interest. A causal model may be partial, explaining only certain elements or aspects of the phenomenon of interest.

The technological rule, on the other hand, is typically studied within its intended context of application, in order to be as sure as possible of its effectiveness, also under the influence of less well-known factors. Grounding a technological rule on explanatory laws does not necessarily mean that every aspect of it (and of its relations with the context) is understood. Typically, several aspects keep their 'black box' character and testing within the context is still very necessary to account for its effectiveness.[8]

The typical research design to study and test technological rules is the multiple case: a series of problems of the same class is solved, each by applying the problem solving cycle. Design knowledge is built up through the reflective cycle (Van Aken, 1994): choosing a case, planning and implementing interventions (on the basis of the problem solving cycle), reflecting on the results and developing design knowledge to be tested and refined in subsequent cases.

In developing and testing a technological rule through the multiple case and in analysing its effectiveness through the cross-case analysis during the reflective cycle, one can gain insight in the indications and contra-indications for the application of that rule and hence also in its application-domain. A technological rule is typically not totally general, but applicable to a certain application-domain, a class of problems.

By borrowing concepts from software development (see e.g. Dolan and Matthews, 1993) one can say that research on technological rules typically goes through a stage of *α-testing*, i.e. testing and further development by the originator of the rule, to be followed by a stage of *β-testing*, i.e. the testing of the rule by third parties.

TECHNOLOGICAL RULES IN THE FIELD OF MANAGEMENT

Academic research in design sciences combines description-driven and prescription-driven research. However, the paradigm of these sciences holds that their ultimate goal is the development of tested and grounded technological rules to be used by the professionals in their field. In our field, however, scientization has produced a strong bias towards description-driven research, even to the extent that many feel that that is the only type of research that deserves academic respectability. This has led Miner (1984, p. 304) to suggest to drop the term 'management' and to call the field 'Organization Science'.

While in the last decades description-driven research has not only enhanced the academic prestige of our field but also has produced impressive results, relevant

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through their potential for conceptual use, I feel that the still pressing relevance problem can be much mitigated by complementing such description-driven research with prescription-driven research. To this end I propose to use Miner's suggestion by making a distinction between *Organization Theory* and *Management Theory*. Organization Theory, then, is produced by research on the basis of the paradigm of the exploratory sciences and Management Theory by research on the basis of the design sciences. Organization Theory can be used in a conceptual way by practitioners and can also be used to feed research in Management Theory. As already proposed by Tsang (1997, pp. 85–6) – without using these terms – Organization Theory results can be used to derive potential technological rules, to be subsequently tested and further developed by Management Theory research and Organization Theory results can also be used as input to ground technological rules.

Technological Rules Derived from Description-driven Research

Examples of technological rules, derived from basically description-driven research are the following:

- If you want to realize a large-scale, complex strategic change, use a process of logical incrementalism (Quinn, 1980).
- If you want effective realization of the outcomes of strategic decision-making, promote perceived procedural fairness (Korsgaard et al., 1995) and active participation of middle management (Woolridge and Floyd, 1990).
- If you want to manage the activities within the operational core of a professional organization, use standardization of skills rather than direct supervision (Mintzberg, 1983).

However, the original research results were not formulated as technological rules. In description-driven research, management implications tend to be treated more or less as an afterthought of the analysis and are not tested as such. Such research uses the perspective of an observer and operates in hindsight. Management Theory research, however, uses the perspective of a player and uses *in prevision* intervention-outcome logic: what intervention should a player use in the given context to realize the desired outcome. Therefore, as will be discussed below, a key element of the research strategies in Management Theory is the in prevision field-testing of technological rules.

Research Products, Based on the Paradigm of the Design Sciences

There are significant differences between the causal models of description-driven research and the technological rules of prescription-driven research. Their causal

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logic is comparable: one or more dependent variables are produced, deterministically or stochastically, through one or more independent ones.[9] However, one difference lies in the nature of the independent variables: in the case of the causal model these are elements already present in reality (and not always manipulable), while in the case of the technological rule it is a designed intervention to solve an improvement problem or a designed artefact, like an organization structure or management system, to solve a construction problem. Or in the words of Cheng and McKinley (1983), in prescription-driven research the independent variables should be *applicable*.

Often the dependent variables are also different. For causal models in the field of management the 'bottom line' (or organizational effectiveness) is a much used result variable (see e.g. Lewin and Minton, 1986; March and Sutton, 1997), or it is even assumed that relevant management research should always focus on overall organizational performance as dependent variable (Cheng and McKinley, 1983, p. 98). However, March and Sutton show that the mechanisms causing variation in overall organizational performance are unstable. To this one may add that overall performance is typically not only influenced by the independent variables of interest, but by many more organizational and contextual variables as well. So the impact of the independent variables tends to 'drown in noise', forcing the researcher to restrict himself or herself to study only those independent variables that have a really strong impact on the bottom line. On the other hand, when testing technological rules one tends to investigate short causal chains. In such testing, the dependent variable is not the ultimate overall organizational performance, but rather one or more operational variables, like the question whether an intended change is realized or not (while it is still assumed that such operational outcomes will eventually contribute to organizational performance).

Causal models can be and often are partial and so explain only certain aspects of the phenomenon of interest. If they are quantitative, they tend also to be strongly reductionistic, forced by the need for quantification. Technological rules, on the other hand, are holistic. A given intervention is applied in a certain context and all organizational and contextual factors have an impact on its outcome. Some of the mechanisms determining its effectiveness will be analysed to ground the technological rule, but other factors will retain their 'black box' character. The description of rule, context and outcome need not be reductionistic, but can use 'thick' qualitative text (Geertz, 1973).

Research Strategies, Producing Technological Rules

Research producing field-tested and grounded technological rules differs in several respects from description-driven research. Prescription-driven research is solution-focused, rather the problem-focused and, as said, it takes the perspective of the player, rather then of the observer. As in other design sciences, the typical research

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strategy in Management Theory is the multiple case-study, using the reflective cycle discussed above (see Eisenhardt (1989) and Parkhe (1993) on the power of the multiple case-study).

There are two types of multiple-case studies in Management Theory, viz. the *extracting* and the *developing* multiple case-study. The extracting multiple case-study is a kind of best-practice research and is aimed at uncovering technological rules as already used in practice. A good example of such research is the classical study of Womack et al. of the automotive industry and especially of Japanese industrial practices (Womack et al., 1990). This research has produced, among other things, a number of very powerful technological rules, like the Kanban-system and Just-in-Time delivery for driving a supply chain.

In the developing multiple case-study the technological rules are developed and tested by the researcher(s) in close collaboration with the people in the field (see e.g. Keizer et al., 2002: the risk diagnosing system they analyse, is the result of a developing multiple case-study). It is clinical research, i.e. the rules are developed and tested in the context of application. Such research is initiated by the researcher(s) interested in developing technological rules for a certain type of managerial issue. Each individual case is primarily oriented at solving the local problem in close collaboration with the local people. But, following the reflective cycle, after each case the researcher develops knowledge that can be transferred to similar contexts on the basis of reflection and cross-case analyses (see Eisenhardt (1989, 1991) for a good description of such an approach).

This development process can first go through a stage of 'α-testing', i.e. analysis of the effectiveness of a certain rule in the original context (like Quinn's (1980) research into logical incrementalism). But invaluable insight can be gained by subsequent 'β-testing' (see Dolan and Matthews, 1993), i.e. translating the rule to other contexts, having third parties use it, assess its effectiveness and make final improvements. One might compare β-testing with replication research, advocated by Tsang and Kwan (1999) for description-driven management research. It is this β-testing, which can provide further insight into the indications and contra-indications for the rule and hence in its application-domain. In principle, much material is already available from some form of testing, since several top academic researchers also have consultancy practices. These practices are, in essence, the translation of their research results to other contexts and both the successes, and especially the less than successful applications, should provide much insight. Unfortunately, such material is too little published. Nevertheless, such testing is still not full β-testing. An essential element of β-testing is that testing is conducted by a third party to counteract the 'unrecognized defences' of the originator of the rule, which may blind him or her to possible flaws in its use (Argyris, 1996).

In description-driven field research the entry into target organizations is sometimes a problem, as the members of those organizations will not always see what

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valuable returns they might get from the time and effort they are expected to invest in such research. In prescription-driven research one has to find organizations that are interested in the problem in question, but – once the researcher has established a basic credibility – entry is much less difficult as there is a win-win situation for both parties. The members of the target organization are often eager to get an external view on their problems and to use the insights from similar cases.

Collaborative Clinical Research

Both the extracting multiple-case study and the developing one are clinical, and the developing one is also strongly collaborative. Collaborative clinical research is, of course, not a new research strategy. Collaborative research is, for instance, discussed in a recent special of the *Academy of Management Journal* (editorial by Rynes et al., 2001).

The clinical research in each individual case is aptly described by Schein (1987), but his cases are client initiated and are less clearly focused on developing explicit, transferable knowledge through a multiple-case strategy. In Schein (2001) he also discusses researcher initiated inquiry, but also in this article the research focus remains somewhat diffuse (Schein, 2001, p. 235). Mode 2 knowledge production, as discussed by Gibbons et al. (1994), more closely resembles Management Theory research as discussed here, although here also the strong emphasis on local knowledge production seems to undermine the idea of developing transferable knowledge. A research focus on the development of explicit, transferable knowledge through a multiple-case strategy can be more clearly found in Eisenhardt (1989, 1991), even if it doesn't become clear from her rebuttal to her critics (Eisenhardt, 1991), whether she aims at reductionistic theory building or at 'thickly' formulated actionable knowledge.

Finally, a clinical approach, intensive interaction with local people and a strong emphasis on local problem-solving can also be found in Action Research (see e.g. Argyris et al., 1985; Clark, 1972; Eden and Huxham, 1996; Reason and Bradbury, 2001; Susman and Evered, 1978). The term Action Research covers a large variety of approaches (see e.g. Dickens and Watkins, 1999, p. 127; Eden and Huxham, 1996, p. 527; Reason and Bradbury, 2001, p. xxiv), but by and large they have the above-mentioned three characteristics in common with Management Theory research, as described here. One of the differences can be the research focus, as several Action Research approaches do not explicitly aim at developing valid knowledge that can be transferred to other contexts. Another difference can be a difference in ethos. Action research is in many cases based on the ethos of industrial democracy, aiming to liberate the weak from the dominant, or to free organization man from the constraints of Weber's 'iron cage' or at 'human flourishing', as Reason and Bradbury (2001, p. 2) put it. The ethos of Management Theory research, as discussed here, is the ethos of the Business School as

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Professional School, fulfilling its mission of developing valid and relevant knowledge that practitioners can use to further the success (or the 'flourishing') of their organizations in the interests of all stakeholders.

Management Theory research as described here may have most in common with the approach to Action Research discussed by Eden and Huxham (1996), who stress among other things the need for generality of research results (p. 530) and write about 'action research aimed at the study of organization and organizations . . . where it is likely that the researcher accepts the dominant managerial ideology' (p. 529).

Grounding on Generative Mechanisms

In Engineering and in Medicine technological rules often can be grounded on the general laws and other results from the natural sciences. Although the social sciences provide vital inputs for our field, the grounding of the technological rules of Management Theory can use such inputs at best in a conceptual way. A more direct way to ground technological rules can be found in the work of Pawson and Tilly (1997).

The field-testing of managerial technological rules has much in common with evaluation research of social programmes, like crime prevention or schooling programmes (see e.g. Cook and Campbell, 1979; Guba and Lincoln, 1989; Pawson and Tilly, 1997). As in Management Theory research such evaluation research follows intervention-outcome logic and is by its very nature testing-in-context. Following Pawson and Tilly (1997), the key question is not so much whether 'it' works, but what it is about the programme that makes it work in terms of generative mechanisms. Their starting point is what they call the basic realist formula: *mechanism + context = outcome*. These mechanisms may include both impersonal, material factors and personal interpretations and discourses of the actors involved. Such testing does not make a paradigmatic choice between a structure and an agency-view. Pawson and Tilly use the example of the introduction of a closed-circuit TV system in a car park to illustrate their use of the concept of generative mechanisms. In this illustration they discuss eight potential mechanisms linking the intervention (the introduction of CCTV) with the desired outcome (reduced crime), ranging from offenders detected by the system and subsequently arrested, punished and deterred, and effective deployment of security staff (towards areas where suspect behaviour is detected), to deterrence of potential offenders due to the increased risk of arrest and more cautious behaviour of car owners (locking their car, removing items from sight due to the fact that a well-publicized CCTV-system reminds them of the risk of theft) (see Pawson and Tilly, 1997, pp. 78–9). The testing of this intervention (the introduction of CCTV) in various settings can give insight in the actual power of each potential mechanism and can lead to design-changes in the intervention.

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A managerial technological rule will usually not be grounded in terms of general laws, as may be the case in Engineering, but rather in terms of generative mechanisms as discussed above. An example of the use of impersonal, immaterial, factors as generative mechanism is Goldratt's Theory-of-Constraints (Goldratt and Cox, 1986). The rule is that in managing a factory one should focus on optimizing the use of the constraining capacity group. The generative mechanism is that it is this group that determines the output of the factory as a whole.

An example of the use of more personal, sense-making factors can be found in Tichy's TPC-model (Tichy, 1983). One rule is that, if a given strategic change hurts the real interests of a certain subgroup, one should use Political interventions rather than Technical or Cultural ones. The generative mechanism is that Technical, i.e. content-oriented interventions, will demonstrate even more clearly to that group that their interests will be hurt, which will not help to overcome their resistance to the change; that Cultural interventions, i.e. inviting participation, will give them the opportunity to organise a coalition against the change; while Political, i.e. power interventions, can be accepted as being the duty of top management to act in the interests of the organization as a whole.

In β-testing of managerial technological rules one is interested in both driving and blocking mechanisms (instances where the rule fails are also highly interesting). It is especially this grounding in driving and blocking mechanisms which will support the translation of the rule to other contexts.

A key criterion for distinguishing academic research results from the prescriptions found in 'Heathrow-literature', is justification. The effectiveness of an algorithmic technological rule (applied as recipe) can be proven[7] in deterministic or stochastic terms. But the indeterminate nature of heuristic rules – and most technological rules in the field of management will be heuristic – makes it impossible to provide such proof. However, through multiple case-studies one can accumulate supporting evidence which can continue until 'theoretical saturation' (Eisenhardt, 1989) has been obtained.

In the case of algorithmic rules, the evidence can be left out after it has been assessed. Application and further research can be based on the rules themselves. In the case of heuristic rules, the evidence remains part of the results. In order to use the results for application or for further research, one keeps needing the evidence – either as it is or in condensed form – to forecast the effectiveness of the application and for translation into the new context.

Differences between Organization Theory and Management Theory

In this article I argue that the relevance problem of academic management research can be mitigated if description-driven research, resulting in what may be called Organization Theory, should be complemented with prescription-driven research, resulting in what may be called Management Theory. This argument is

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Table I. The main differences between description-driven and prescription-driven research programmes

Characteristic	*Description-driven research programmes*	*Prescription-driven research programmes*
Dominant paradigm	Explanatory sciences	Design sciences
Focus	Problem focused	Solution focused
Perspective	Observer	Player
Logic	Hindsight	Intervention-outcome
Typical research question	Explanation	Alternative solutions for a class of problems
Typical research product	Causal model; quantitative law	Tested and grounded technological rule
Nature of research product	Algorithm	Heuristic
Justification	Proof	Saturated evidence
Type of resulting theory	Organization Theory	Management Theory

summarized in Table I, showing the main differences – more or less in black and white – between description-driven and prescription-driven research programmes in our field.

Description and Prescription in Functional Management Fields

The tensions between description-driven and prescription-driven research and between the academic and the professional reputation systems do not only exist in the field of organization and management theory in general, but also in functional areas like Operations Management, Management of Technology, Marketing Management and Human Resources Management. The situation in these areas is somewhat different, because here professional associations, which publish journals and organize conferences, bring together practitioners and academics. These conferences tend to suffer less from what Hambrick calls the 'incestuous, closed loop' of the American Academy of Management conferences (Hambrick, 1994, p. 13). Furthermore, in these fields one finds more academics with mixed academic/professional backgrounds as opposed to academics with purely academic backgrounds. This causes some more interest in the utility of research products and in the professional reputation system.

However, the forces of the academic reputation system are also felt strongly in these fields as illustrated by the following two examples. In the field of Operations Management, Meredith et al. (1989) bemoan the dominance in their field of Operations Research – high in academic prestige – at the expense of research products more relevant for real-life problems (see also Whitley (1988) on the clash between the two reputation systems in this field). In Management of Technology, Ottosson

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castigates the 'terror of statistical investigations', 'concentrating on measurable numbers to get nice tables processed with advanced computer programs', which are 'well received in scientific society', but leave 'managers and entrepreneurs, seeking useful theories for their every-day businesses' in the cold (Ottosson, 1998, p. 236).

Nevertheless, because of its more instrumental nature, the literature in these functional fields *does* give practitioners, like marketing managers and logistic managers, essential knowledge without which they would perform suboptimally. One may compare this with our field of management-in-general, which as yet provides fairly little instrumental knowledge (exceptions being e.g. psychological theories on motivation, which score high on both utility and validity according to Miner (1984), and research on organizational change management, which often has a strong instrumental component).

THE UTILITY OF TECHNOLOGICAL RULES IN THE FIELD OF MANAGEMENT

The quest for prescriptions in the field of management is as old as the field itself and has not been invented here. It is older than the tradition of description-driven research based on the trinity of description, explanation and prediction. My proposition, therefore, is not the invention of the idea of prescription, but to take prescription-driven management research serious academically by rigorous testing and grounding.

By their very nature technological rules are much more application-oriented than the causal models of description-driven research. I discuss here the utility of technological rules by examining the extent to which they fulfil the five key user-needs of practitioners regarding (academic) management theory as developed by Thomas and Tymon (1982) on the basis of various criticisms of such theory (see also Cheng and McKinley (1983) on the requirements of relevant management theory).

- *Descriptive relevance* or external validity: the raison d'être of a technological rule is its external validity as established by testing in multiple case-studies as discussed above.
- *Goal relevance* or the extent to which research results refer to matters the practitioner wishes to influence: in a prescription-driven research programme goal relevance is a key criterion for the choice of rules to be developed, tested and grounded.
- *Operational validity* or the extent to which the practitioner is able to control the independent variables in the model: the very nature of a technological rule assures its operational validity.

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- *Non-obviousness*: because a technological rule is not forced into a reductionistic format as quantitative causal models are, there is little danger of overly obvious research results.
- *Timeliness*: a practitioner need arising from the 'incredible long periods of time' required to adequately assess organizational phenomena and the scientists' reluctance to make recommendations before all the facts are in (Thomas and Tymon, 1982, p. 349): in this respect the technological rule has no advantage over the causal model; for classes of management problems for which timeliness is a real issue, the practitioner will have to deal with consultants rather than with academic researchers.

This brings me to the overall conclusion that on the basis of the Thomas and Tymon criteria, generally speaking, the utility of the technological rule is significantly higher than that of the traditional causal model.

DISCUSSION

Discussions on the use of the paradigm of the design sciences in management research and on the quest for field-tested and grounded technological rules, have shown several areas of possible criticisms. Of these I will discuss four.

In the first place it is often implied, both by advocates and critics of more instrumental Management Theory, that such theory should have a general applicability and/or should be specific enough to be used as a kind of recipe. However, technological rules are not general knowledge, like e.g. Maslow's hierarchy of needs, but rather mid-range theories of practice. They are only valid for a certain application domain, a range of settings that have key attributes in common with the settings in which the rules were developed and tested (Kennedy, 1979).[10] Nor are technological rules to be used as recipes. Although a technological rule is to be used in an instrumental way, it is no specific prescription for a specific setting. A practitioner has to design his/her own intervention or system, on the one hand based on his/her experience, creativity and deep understanding of his/her local setting, and on the other on the knowledge of the appropriate technological rules, of their generative mechanisms and of their supporting evidence.

In the second place, the development of more instrumental Management Theory might be regarded as privileging only one group of organizational stakeholders, i.e. managers. The position from which this article is written, is that people join organizations, work for organizations and are often very committed to their organizations in the expectation that the success of their organization is not only good for owners and managers, but also for themselves and for their peers. Developing theory, the application of which may contribute to organizational success is in principle not only in the interest of managers but of all stakeholders. By analogy with McGregor's Theory X and Y with respect to views on people, one might call

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this a kind of Theory Y on managers and management. Critical management theory (see e.g. Alvesson and Deetz, 2000; Grey and Mitev, 1995), then, might be called a kind of Theory X on managers: there is a deep divide between management and the worker and supporting management is inherently detrimental to the worker. In Theory X the development of instrumental Management Theory privileges managers, in Theory Y it can be in the interest of all stakeholders. Using a Theory Y on managers, as I do here, does not imply that one should close one's eyes for possible conflicts of interest between organizational stakeholders, for possible abuse of power by managers or for the instances of top management greed and corruption one sees today. However, similar issues tend to be present in every field of human endeavour and can in principle be dealt with here as elsewhere.

In the third place my use of analogies between management on the one hand and Engineering and Medicine on the other is not new (see e.g. Squires, 2001; Whitley, 1988) and is not meant to prove anything. I use the analogy to show the differences between the paradigm of the design sciences with that of the explanatory sciences and to show that research on the basis of the paradigm of the design sciences should not necessarily result in instrumentalism (i.e. working with theoretically ungrounded rules of the thumb; Archer, 1995, p. 153) or in turning the field again into a practice-based craft, but can indeed deserve academic respectability.

On the other hand the analogy has its dangers, as design-oriented research in management is in several respects more complex than design-orientated research in Engineering and Medicine. The reasons for that include the fact that management tends to be more context-bound than these other disciplines, that the number of cases to base generalizations on is much lower and that – as will be discussed below – the generative mechanisms linking intervention with outcome are often of an immaterial nature. Because of this, I do not expect that the results of Management Theory research will obtain the privileged status of the foremost source of knowledge for practitioners as is the case in Engineering and Medicine. Although in the heady days of the scientization of Business Schools, Andrews (1969) did argue that management should be regarded as a profession, there are also reasons not to do so (see e.g. Raelin, 1990).[11] An important one is that formal managerial knowledge, i.e. formal knowledge on management-in-general, is not as central to managerial success as such knowledge is in 'strong' professions. Managers will continue to be inspired by a range of sources of knowledge. However, in my opinion, academic management research should at least strive to be one of the more important of these sources.

Several authors, like Koontz (1961, 1980) and Pfeffer (1993), argue that the variety of approaches advocated in organization and management literature is a sign of immaturity of the field. Indeed, the level of 'paradigm development' (Cheng and McKinley, 1983) of our field is still fairly low. However, variety can be a source of inspiration for practitioners and can, in my opinion, also be seen

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as an expression of the richness of our field. Besides, it is doubtful whether practitioners expect the field to develop one Grand and Unified Management Theory. What *is* possibly a sign of weakness is the plethora of *unopposed* so-called management fads (Byrne, 1986; Pascale, 1990, p. 20). It could be a rewarding task for prescription-driven research to do some weeding here with the help of some rigorous testing and grounding.

And, finally, there is the slippery ground of the paradigm discussions. As said earlier, in this article I have used the term 'paradigm' to denote the combination of research questions asked, research methodologies allowed to answer them and the nature of the pursued research products. And I have used this concept to discuss and analyse the differences between explanatory and design sciences and have proposed to create for our field space for both Organization Theory research, based on the paradigm of the explanatory sciences, and for Management Theory research, based on the paradigm of the design sciences. However, this does not necessarily mean that these two types of research programmes would be incommensurable with respect to their ontological, epistemological and even with respect to many of their methodological starting points. Like Schulz and Hatch (1996), I would rather stay out of paradigm wars.

Nevertheless, as Archer (1995, p. 2) says, 'what social reality is held to be cannot but influence how society is studied'. Likewise, what I hold management to be, influences how I study it. Therefore, I will discuss the ontological and epistemological starting points on which my discussion on research products and research strategies is based. This discussion, however, is *not* meant to exclude other possible ontological and epistemological starting points to drive respectively description-oriented Organization Theory research and prescription-oriented Management Theory research.

In this article the discussion on management and organization, on technological rules and on the generative mechanisms, linking intervention with outcome, is based on realism (see e.g. Archer, 1995; Sayer, 1992). I follow realism's contention that there exists a real (material) world, independent from observers and their knowledge. We can develop knowledge of that real world through our senses, even though sensory experiences are concept-laden and are therefore no objective images of the external world. Objects in the real world have particular causal powers that can produce under specific conditions certain effects.

A similar line of reasoning applies to the (more immaterial) social world: it exists ontologically independent of the observer, even though social phenomena themselves are ontologically concept-dependent. We can develop knowledge on the social world through sensory experiences and the social world has 'structure' that influences and constrains 'agency' (while 'agency' subsequently creates 'structure'; Archer, 1995).

The application of these ontological and epistemological starting points to management gives the following line of reasoning. Like Mary Parker Follett, I see man-

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agement as the art of getting things done by others. Hence management is about communication with others in the immaterial domain of language games (Wittgenstein, 1953) in order to mobilize those others to act in the material domain of action to produce as good as possible intended material outcomes. Technological rules are largely about communication interventions to produce eventually material outcomes. Research in Management Theory is aimed at developing sound technological rules and at uncovering the generative mechanisms that link (immaterial) intervention with (material) outcomes. As discussed, such generative mechanisms can be of a material nature, but most are of an immaterial, sense-making nature.

Furthermore, the design and implementation of management interventions and systems is meant to change organizations in a some respect in order to improve performance. Like other social institutions, organizations are ontologically social constructions (Berger and Luckman, 1967; Searle, 1995). And from a design perspective organizations are at the same time artefacts, created under the influence of design-oriented human actions, and natural systems, developing under the influence of the interactions and the self-organization and self-control of their stakeholders. In this respect an organization can be compared with a garden, as artefact created through the designs and hard work of the gardener and as natural system developing under the influence of sun, rain, soil conditions, insects etc. (and in some gardens the gardener tries to control natural development as best as he/she can, while in others the gardener leaves more to Mother Nature).

CONCLUSION

The main thesis of this article is that the relevance problem of academic management research can be mitigated if one would create space for Management Theory research, based on the paradigm of the design sciences, next to the more traditional Organization Theory research, based on the paradigm of the explanatory sciences. As said, both research programmes can well operate in a profitable partnership. Organization Theory could provide understanding of the problem as a basis for developing technological rules and insight in causal mechanisms as a basis to uncover generative mechanisms, while Management Theory could provide further insight into the nature of managerial processes and generating new research questions. Such an approach would meet Dewey's (1929) criticisms on the traditional separation of knowledge and action and follow Starbuck and Nyström's (1981) adage, 'if you want to understand a system, try to change it'.

An increasing interest in prescription-driven management research could lead – in terms of McKinley et al. – to a certain 'school of thought', i.e. an integrated theoretical framework that provides a distinct view on organizations and that is associated with an active stream of empirical research (McKinley et al., 1999, p. 635). However, a more ambitious result would be an effective partnership of

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description-driven and prescription-driven research in many schools of thought, thus giving our field the character of a technology more than that of a basic science, or – in terms of Gibbons et al. (1994) – increasing the use of mode 2 knowledge production in our field.

Whether that will happen, will ultimately be a matter of values. Academics may consider it unacademic to be much concerned with praxis, rather like Roman senators who were not supposed to be involved in craft or trade. A quest for field-tested and grounded technological rules, which in the field of management will be predominantly qualitative and heuristic by nature, means trading the priestly beauty of truth for the soldiery glory of performance (to paraphrase March and Sutton, 1997) and that may be too high a price. Some may fear that a stronger praxis-orientation will cause the field to relapse into a 'practice-based craft'. In medieval times, the medical doctor did not soil his hands, but left the butchery to the chirurgeon. In modern times, this barber-surgeon emancipated and became an academically respectable surgeon. Ultimately, I expect this to happen to utility-conscious academic management researchers too, as long as they focus on rigorous testing and grounding of their technological rules.

NOTES

[1] In this article I use the term 'scientific' like the German 'wissenschaftlich' or the Dutch 'wetenschappelijk', meaning 'according to sound academic standards'; thus, its meaning is not confined to the natural sciences.

[2] Some authors, like Child (1995) and Koza and Thoenig (1995), suggest that in our field one might contrast a European tradition, which is concerned more with the academic reputation system, with an American one, which is concerned more with the professional reputation system. This may be true for European scholars with a background predominantly in sociology, but not necessarily so for organization and management scholars with a different background.

[3] In a discussion on the quality of research products in the field of strategic management, Montgomery et al. (1989, p. 191) admit that 'sciences should be undertaken for the sake of ultimate application', but then proceed to stress the 'ultimate' in their statement, urging editors of academic journals not to demand *direct* application of the research products presented (as if they always do), thus leaving more room for the fine tuning of (descriptive) theory. In a reaction Seth and Zinkhan (1991) take it a step further and complain that Montgomery et al. are too application-oriented [sic] and propose that 'if strategic management is to become a science it must strive towards "explaining by law" the phenomena of interest' (Seth and Zinkhan, 1991, p. 80), thus forcing their causal models into a reductionistic, quantitative format.

[4] Pelz (1978) also discussed a third type of use, viz. *symbolic* use of scientific knowledge: the use of that knowledge to legitimate predetermined positions.

[5] In the context of the design sciences, Simon primarily discusses construction problems, while in this article the design sciences may also deal with improvement problems.

[6] It is the mission of Design Research and Design Theory (see, e.g. Cross, 1993; Evbuomwan et al., 1996; Hubka and Eder, 1996) to contribute to the process knowledge of the designer. However, the gap between design theory and design practice is at least as large as the gap between management theory and practice (see e.g. Dorst, 1997; Norman, 1996; Van Handenhoven and Trassaert, 1999). It is with respect to *object knowledge* that (academic) research in the design sciences is so successful, not with respect to process knowledge.

[7] Conclusive proof, of course, is only possible in the formal sciences. In this context 'proof' should be read as 'convincing internal validity', and will be contrasted with the weaker form of internal validity for heuristic technological rules, i.e. supporting evidence.

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[8] This is not to say that *all* technological rules have to be tested within their context. In the engineering sciences in particular, it is possible to isolate certain research subjects from their context without losing essential characteristics.

[9] Or – if one wants to use a more sophisticated theory of causation (Sayer, 1992, pp. 103–16) – certain objects have causal powers than can produce under specific conditions certain events.

[10] An example of the problems of transferring knowledge from one context to another can be found in Weick and Sutcliffe (2001), who give a very insightful analysis of what they call 'High Reliability Organizations' on the basis of case studies of, among others, the operations of an aircraft carrier and of a nuclear plant. However, in their discussion of the management implications of their results they tend to generalize far beyond the settings in which their insights were developed. The careful definition of its application domain is a key element of technological rule research.

[11] Or see the balanced discussion of this issue by Squires (2001, pp. 483–5), who concludes that management may not be a 'strong' profession, but still has several important characteristics of a profession.

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